

1 A Technical Appendices and Supplementary Material

2 In this section, we explore the power law symmetry experiment in greater detail. Specifically, we
 3 wish to compare the speed at which the three methods increase in accuracy, both as a function of
 4 epochs and as a function of training time. Therefore, the power law symmetry experiment is repeated,
 5 and each method is trained for 12 epochs. Model accuracies are recorded for each epoch, and 10
 6 trials for each method are conducted. So that a more realistic timing scenario can be realized, GPU is
 7 used. For the regularization method, the first epoch is trained with the regularization parameter set to
 8 0, which is equivalent to the baseline model.

9 We estimate that the median time for each epoch in the Baseline case—that is, with neither augmenta-
 10 tion nor symmetry regularization—is approximately 16.5840 seconds. In the case of the augmented
 11 method, in which training data is effectively tripled, the median epoch time is 18.0781 seconds.
 12 Lastly, the symmetry-regularized method has a median epoch time of 18.0710 seconds. Thus, when
 13 comparing the augmentation approach and the symmetry-regularized approach, we consider the
 14 accuracy as a function of epochs to also describe accuracy as a function of time.

15 First, we present a graphical representation of the median accuracies for each of the three methods at
 16 the conclusion of the final epoch. The error bars represent half of the IQR. The visual is given in
 17 Figure 1.

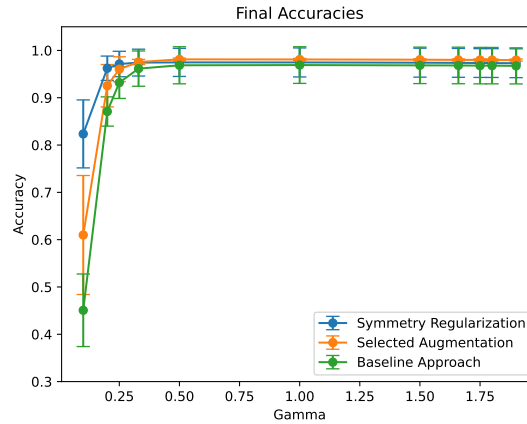


Figure 1: Final Accuracies for each method for the chosen values of γ .

18 Evident from Figure 1 is the fact that the accuracies for each method for all but the lowest two values
 19 of γ , namely 0.1 and 0.2, are statistically tied. For $\gamma = 0.1$, our method outperforms the others,
 20 which are statistically tied. For $\gamma = 0.2$, the baseline method underperforms when compared to the
 21 other methods, which other methods are tied. However, Figure 2 depicts accuracies at the penultimate
 22 epoch, offering additional insight.

23 Figure 2 reveals that, at the penultimate epoch, the symmetry regularization term uniformly outper-
 24 forms the baseline method. The augmentation method outperforms the baseline method except at
 25 the lowest two values of γ . This suggests that the augmented and regularized approaches may learn
 26 faster than the baseline approach.

27 We also consider the median accuracies at the midpoint of training, which is epoch 6. Figure 3 depicts
 28 these accuracy scores.

29 In Figure 3, we see that the regularization method uniformly outperforms both competing methods,
 30 and that the competing methods are statistically tied. This suggests that the regularization method
 31 may learn faster than the competitors. This result is corroborated by Figure 4, which depicts the
 32 median accuracies as a function of the epoch number for $\gamma = 1.0$ (the original test data).

33 In Figure 4, the augmented and baseline methods appear to flatline before encountering a sudden
 34 surge in accuracy later in the training process. Meanwhile, the accuracies for the regularized method

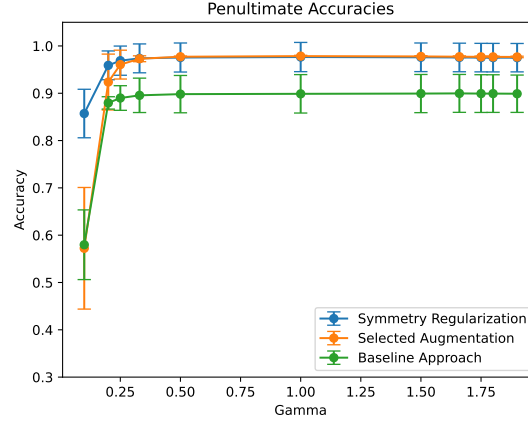


Figure 2: Penultimate Accuracies for each method for the chosen values of γ .

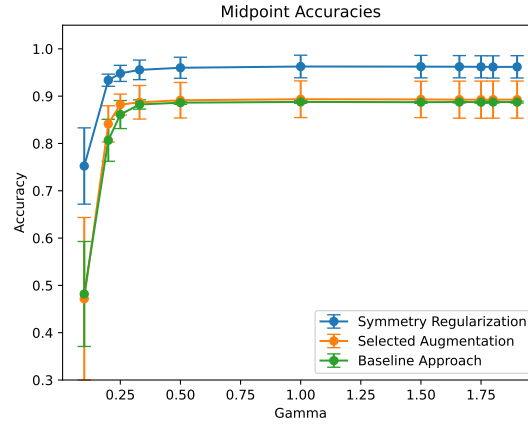


Figure 3: Midpoint accuracies for each method for the chosen values of γ .

35 increase steadily before becoming matched by the other methods. A similar picture holds for $\gamma \geq 0.33$.
 36 For smaller values of γ , however, the picture differs slightly, as is seen in Figure 5.

37 In Figure 5, we see that the regularization method tends to obtain higher accuracy values earlier, as
 38 before. However, we also see that the augmentation and baseline methods are slower to increase in
 39 accuracy when compared with higher values of γ .

40 We believe our experiment suggests that learning occurs faster, initially, using the symmetry-
 41 regularized method. The augmented method appears to obtain high accuracy scores earlier than the
 42 baseline model. One final visual to support this is Figure 6.

43 Figure 6, like Figure 3, shows the regularization method uniformly outperforming the other methods.
 44 Since the time per epoch for the regularization and augmented methods is approximately the same,
 45 it is clear that halfway through the training time, the regularization method was outperforming the
 46 augmentation method. One may naturally consider whether, at this time, the baseline method was
 47 outperformed. Figure 6 demonstrates that the baseline method at the end of epoch 10 was statistically
 48 tied to the augmentation method and uniformly outperformed by the regularization method. Although
 49 the median time for the baseline method was estimated to be lower than for the other methods, 10
 50 epochs for the baseline method still takes significantly more time than 6 epochs for either of the other
 51 methods, by our estimates. Therefore, the regularization method learned more quickly in the early
 52 stages of training than both competing methods.

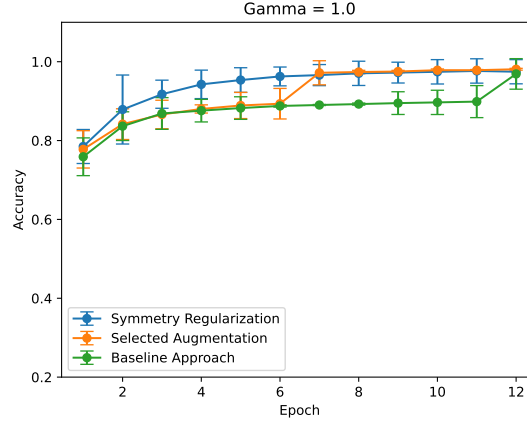
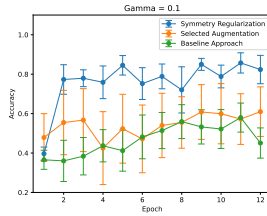
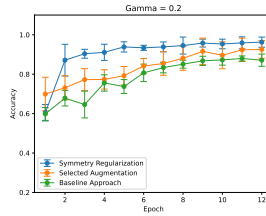


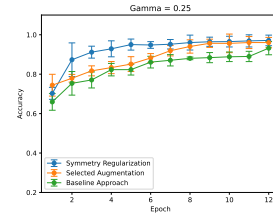
Figure 4: Accuracies for $\gamma = 1.0$ —the original test data—as a function of the epoch number.



(a) The error bars are usually wider for $\gamma = 0.1$.



(b) Regularization uniformly outperforms the baseline for $\gamma = 0.2$.



(c) A more steady increase in accuracies for the augmented method occurs for $\gamma = 0.25$

Figure 5: Accuracy vs. epoch curves for low values of γ . In (a), we see that accuracy scores struggle to increase, but are generally higher for the regularization method. In (b), we see that the augmentation method eventually reaches the accuracy levels of the regularization method, but that the baseline method does not. In (c), the lowest value of γ which appears in the augmentation, the general trend of Figure 4 appears, though the augmentation method does not appear to increase as quickly.

In summary, our experiment suggests that while each method can eventually obtain competitive accuracy scores for most values of the continuous parameter γ , the symmetry regularization approach appears to offer a few advantages. Early in training, this method appears to have a tendency to outperform the competing methods. Additionally, symmetry regularization appears to have had more success at one end of the extreme values of γ —that is, the values extending beyond those used in augmentation.

Symmetry regularization also offers a conceptual advantage. For continuous transformations, it is not possible to augment a training set to account for every possible value of the parameter. Additionally, one may wish to create a classifier that is robust to many types of symmetries, such as affine transformations, power law transformations, reflections, and perhaps other pixel-based transformations. Symmetry regularization, while perhaps not suited for discrete symmetries, may offer a way to alleviate much of the augmentation required to train deep learning models on image data.

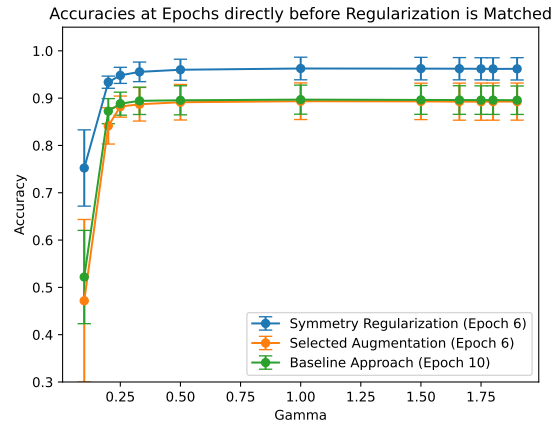


Figure 6: Accuracies at the specified, model-specific epoch as a function of γ .